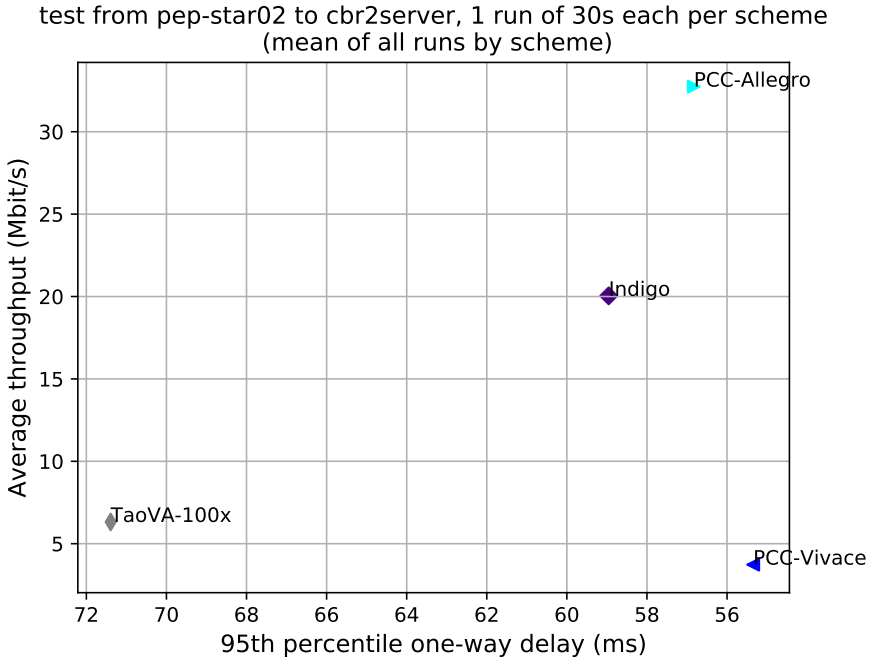
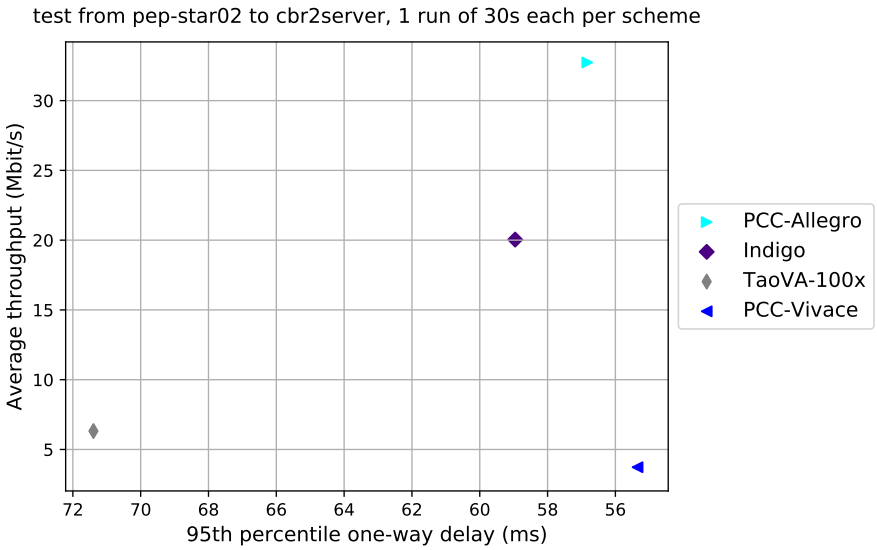
Pantheon Report

Generated at 2026-11-09 23:40:08 (UTC).
Data path: pep-star02 (*local*) →cbr2server (*remote*).
Repeated the test of 4 congestion control schemes once.
Each test lasted for 30 seconds running 1 flow.
NTP offsets were measured against 202.65.114.202 and have been applied to correct the timestamps in logs.

```
System info:
Linux 7.0.0-42-generic
net.core.default_qdisc = fq
net.core.rmem_default = 212992
net.core.rmem_max = 4194304
net.core.wmem_default = 212992
net.core.wmem_max = 4194304
net.ipv4.tcp_rmem = 4096 131072 33554432
net.ipv4.tcp_wmem = 4096 16384 4194304

Git summary:
branch: HEAD @ 23e738ce5acae1d36e321886cd613b0b9401ac11
third_party/genericCC @ d0153f8e594aa89e93b032143cedbdfe58e562f4
third_party/indigo @ 463d89b09699a57bfdfbae351646df6a60040b90
third_party/pantheon-tunnel @ f866d3f58d27afd942717625ee3a354cc2e802bd
third_party/pcc @ 1afc958fa0d66d18b623c091a55fec872b4981e1
 M receiver/src/buffer.h
 M receiver/src/core.cpp
 M sender/src/buffer.h
 M sender/src/core.cpp
third_party/vivace @ 2baf86211435ae071a32f96b7d8c504587f5d7f4
```


test from pep-star02 to cbr2server, 1 run of 30s each per scheme
(mean of all runs by scheme)
PCC-Allegro
Indigo
TaoVA-100x
PCC-Vivace
Average throughput (Mbit/s)
95th percentile one-way delay (ms)
30
25
20
15
10
5
72
70
68
66
64
62
60
58
56


test from pep-star02 to cbr2server, 1 run of 30s each per scheme
Average throughput (Mbit/s)
95th percentile one-way delay (ms)
PCC-Allegro
Indigo
TaoVA-100x
PCC-Vivace
30
25
20
15
10
5
72
70
68
66
64
62
60
58
56

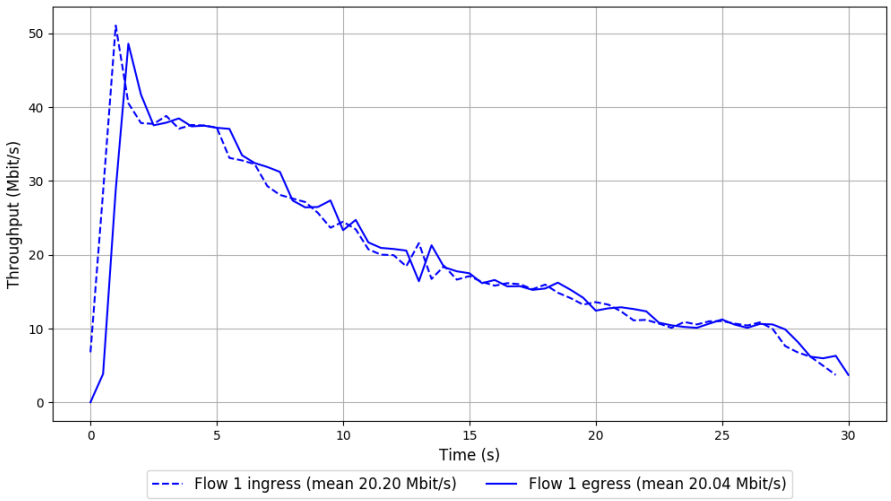
scheme	# runs	mean avg tput (Mbit/s) flow 1	mean 95th-%ile delay (ms) flow 1	mean loss rate (%) flow 1
Indigo	1	20.04	58.96	0.64
PCC-Allegro	1	32.73	56.83	0.48
TaoVA-100x	1	6.32	71.39	0.51
PCC-Vivace	1	3.73	55.35	0.11

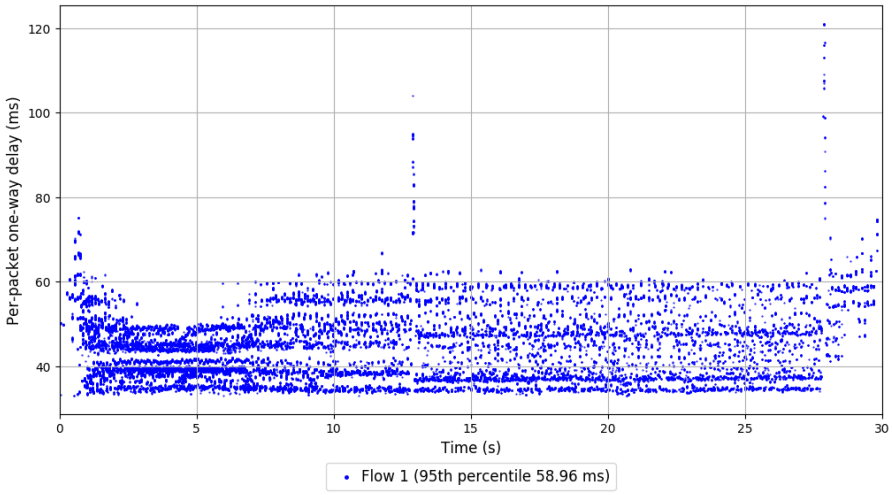
```
Run 1: Statistics of Indigo

Start at: 2026-11-09 22:48:29
End at: 2026-11-09 22:48:59
Local clock offset: 1.957 ms
Remote clock offset: -0.157 ms

# Below is generated by plot.py at 2026-11-09 23:40:07
# Datalink statistics
-- Total of 1 flow:
Average throughput: 20.04 Mbit/s
95th percentile per-packet one-way delay: 58.956 ms
Loss rate: 0.64%
-- Flow 1:
Average throughput: 20.04 Mbit/s
95th percentile per-packet one-way delay: 58.956 ms
Loss rate: 0.64%
```

Run 1: Report of Indigo — Data Link



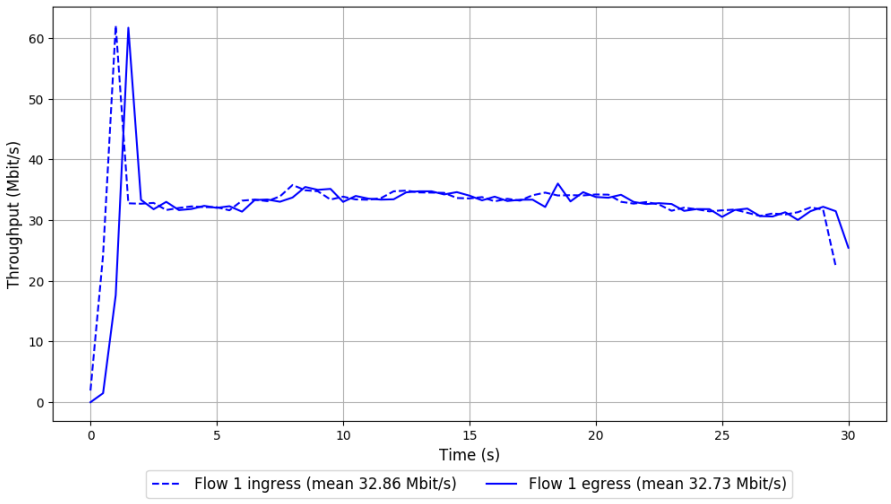


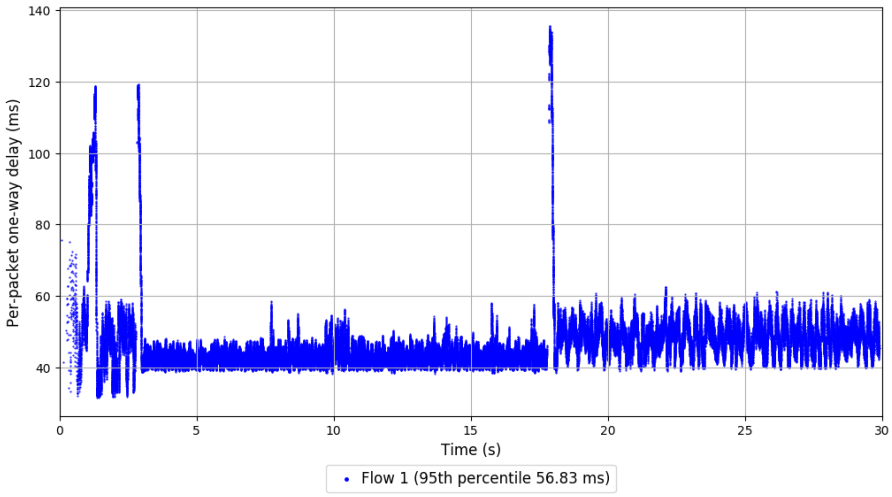
```
Run 1: Statistics of PCC-Allegro

Start at: 2026-11-09 22:49:39
End at: 2026-11-09 22:50:09
Local clock offset: 0.858 ms
Remote clock offset: 0.413 ms

# Below is generated by plot.py at 2026-11-09 23:40:08
# Datalink statistics
-- Total of 1 flow:
Average throughput: 32.73 Mbit/s
95th percentile per-packet one-way delay: 56.829 ms
Loss rate: 0.48%
-- Flow 1:
Average throughput: 32.73 Mbit/s
95th percentile per-packet one-way delay: 56.829 ms
Loss rate: 0.48%
```

Run 1: Report of PCC-Allegro — Data Link



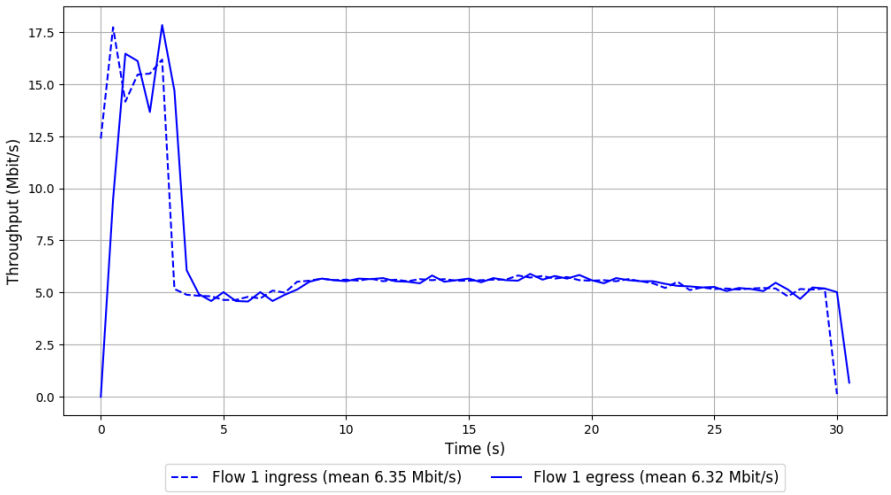


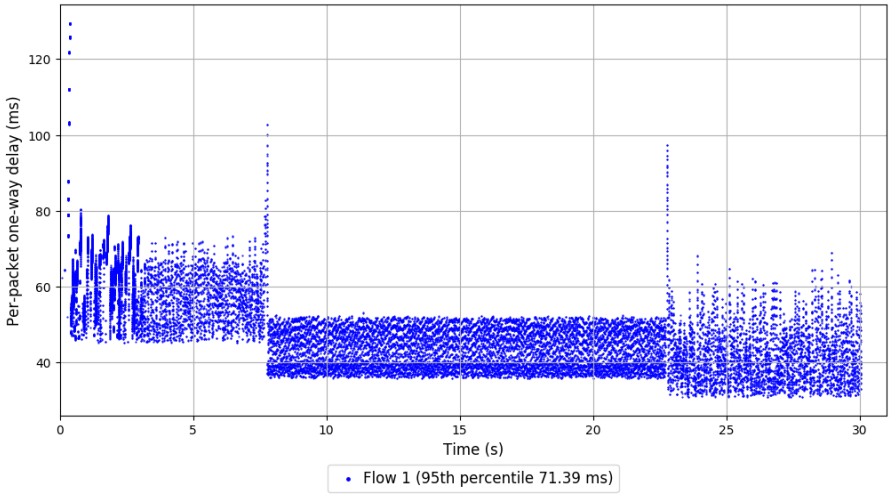
```
Run 1: Statistics of TaoVA-100x

Start at: 2026-11-09 22:50:49
End at: 2026-11-09 22:51:19
Local clock offset: 0.916 ms
Remote clock offset: -1.302 ms

# Below is generated by plot.py at 2026-11-09 23:40:08
# Datalink statistics
-- Total of 1 flow:
Average throughput: 6.32 Mbit/s
95th percentile per-packet one-way delay: 71.393 ms
Loss rate: 0.51%
-- Flow 1:
Average throughput: 6.32 Mbit/s
95th percentile per-packet one-way delay: 71.393 ms
Loss rate: 0.51%
```

Run 1: Report of TaoVA-100x — Data Link





```
Run 1: Statistics of PCC-Vivace

Start at: 2026-11-09 22:51:58
End at: 2026-11-09 22:52:28
Local clock offset: 0.906 ms
Remote clock offset: -0.798 ms

# Below is generated by plot.py at 2026-11-09 23:40:08
# Datalink statistics
-- Total of 1 flow:
Average throughput: 3.73 Mbit/s
95th percentile per-packet one-way delay: 55.347 ms
Loss rate: 0.11%
-- Flow 1:
Average throughput: 3.73 Mbit/s
95th percentile per-packet one-way delay: 55.347 ms
Loss rate: 0.11%
```

Run 1: Report of PCC-Vivace — Data Link

